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Effective Date: 12/10/2023 Target Review Date: 10/10/2026	
Title: Hyperbilirubinaemia	
Jaundice is one of the most common conditions requiring medical attention in newborn babies. Approximately 60% of term and 80% of preterm babies develop jaundice in the first week of life, and about 10% of breastfed babies are still jaundiced at 1 month of age. In most babies with jaundice there is no underlying disease, and this early jaundice (termed ‘physiological jaundice’) is generally harmless. However, there are pathological causes of jaundice in the newborn, which, although rare, need to be detected. Such pathological jaundice may co-exist with physiological jaundice Objective: This guideline focusses on unconjugated hyperbilirubinaemia and its management, Policy Statement: This policy to be used in NICU and any other wards who deal with neonates in this institution. Policy Scope: This is to be used by doctors and nurses in NICU and any other wards who deal with neonates in this institution. 1 1-Information for parents and carers Offer parents or carers information about neonatal jaundice 1.1-Information should include:	

- The importance of recognising jaundice in the first 24 hours and of seeking urgent medical advice
- The importance of checking the baby's nappies for dark urine or pale chalky stools
- The fact that neonatal jaundice is common, and reassurance that it is usually transient and harmless
- Reassurance that breastfeeding can usually continue.

1.2-Information for parents or carers on treatment

Offer parents or carers information about treatment for hyperbilirubinaemia, including:

- Anticipated duration of treatment
- Reassurance that breastfeeding, nappy-changing and cuddles can usually continue.
- Encourage mothers of breastfed babies with jaundice to breastfeed frequently, and to wake the baby for feeds if necessary.

2-Risk factors for hyperbilirubinaemia

Identify babies as being more likely to develop significant hyperbilirubinaemia if they have any of the following factors:

- Gestational age under 38 weeks
- A previous sibling with neonatal jaundice requiring phototherapy
- Mother's intention to breastfeed exclusively
- Visible jaundice in the first 24 hours of life.

Ensure that adequate support is offered to all women who intend to breastfeed exclusively*

3.In all babies:

- check whether there are factors associated with an increased likelihood of developing significant hyperbilirubinaemia soon after birth
- examine the baby for jaundice at every opportunity especially in the first 72 hours.
- Parents, carers and healthcare professionals should all look for jaundice (visual inspection).

3.1-Symptoms and signs:

- When looking for jaundice (visual inspection):
- check the naked baby in bright and preferably natural light
- examination of the sclerae, gums and blanched skin is useful across all skin tones.

Do not rely on visual inspection alone to estimate the bilirubin level in a baby with jaundice.

Do not measure bilirubin levels routinely in babies who are not visibly jaundiced.

3.2-Assess

- Pallor (haemolysis)
- Poor feeding, drowsiness (neurotoxicity)
- Hepatosplenomegaly (blood-group incompatibility or cytomegalovirus)
- Splenomegaly (spherocytosis)

- Physiological
- Prematurity
- Increased bilirubin load:
- blood group incompatibility (Rhesus or ABO)
- G6PD deficiency and other red cell enzyme deficiencies
- congenital spherocytosis
- cephalhaematoma, bruising
- Rarely infection (e.g. UTI, congenital infection)
- Metabolic disorder

4.2-Persistent jaundice after aged 14 days

- Breast milk jaundice
- Hypothyroidism
- Liver disease (e.g. extra hepatic biliary atresia and neonatal hepatitis)
- Alpha-1-antitrypsin deficiency
- Galactosaemia
- TPN-induced cholestasis

5-Investigations

5.1-Assessment of jaundice

- Babies aged <72 hr, at every opportunity (risk factors and visual inspection)
- Babies with suspected or obvious jaundice, measure and record bilirubin level urgently
- <24 hr: within 2 hr
- ≥24 hr: within 6 hr
- If serum bilirubin >100 micromol/L in first 24 hr
- repeat measurement in 6–12 hr
- interpret result in accordance with baby's age and gestation – see Table
- urgent medical review as soon as possible (and within 6 hr)
- Interpret bilirubin levels according to the baby's postnatal age in hours and manage hyperbilirubinaemia according to the (threshold table) and treatment(threshold graphs)

5.2-Jaundice requiring treatment

- Total bilirubin
- Baby's blood group and direct Coombs' test (interpret result taking into account strength of reaction and whether mother received prophylactic anti-D immunoglobulin during pregnancy)
- Mother's blood group and antibody status (should be available from maternal healthcare record)
- PCV

5.3-Plus (if clinically indicated)

- Full infection screen (in an ill baby)
- G6PD level and activity (if indicated by ethnic origin: Mediterranean, Middle Eastern, South East Asian)
- FBC and film

5.4-Persistent jaundice >14 days term baby; >21 days preterm baby

- Total and conjugated bilirubin

- Baby's blood group and direct Coombs' test (interpret result taking into account strength of reaction and whether mother received prophylactic anti-D immunoglobulin during pregnancy)
- Ensure routine metabolic screening performed (including screening for hypothyroidism)
- Urine culture

Baby with conjugated bilirubin >25 micromol/L, refer urgently to a specialist centre

5.5-2nd line investigations

- Liver function tests (ALT, AST, albumin, GGT)
- Coagulation profile
- G6PD screen in African, Asian or Mediterranean babies
- Thyroid function tests: ask for 'FT4 priority and then TSH'
- Congenital infection screen
- Urine for CMV PCR, toxoplasma ISAGA-IgM and throat swab for HSV culture/PCR
- Metabolic investigations e.g:
 - blood galactose-1-phosphate
 - urine for reducing substances
 - alpha-1-antitrypsin

6-How to measure the bilirubin level

6.1-When measuring the bilirubin level:

- use a transcutaneous bilirubinometer in babies with a gestational age of 35 weeks or more and postnatal age of more than 24 hours
- if a transcutaneous bilirubinometer is not available, measure the serum bilirubin
- if a transcutaneous bilirubinometer measurement indicates a bilirubin level greater than 250 micromol/litre check the result by measuring the serum bilirubin
- always use serum bilirubin measurement to determine the bilirubin level in babies with jaundice in the first 24 hours of life
- always use serum bilirubin measurement to determine the bilirubin level in babies less than 35 weeks gestational age
- always use serum bilirubin measurement for babies at or above the relevant treatment threshold for their postnatal age, and for all subsequent measurements
- Do not use an icterometer.

7-How to manage hyperbilirubinaemia

- Use the bilirubin level to determine the management of hyperbilirubinaemia in all babies (see threshold graphs).
- Do not subtract conjugated bilirubin from total serum bilirubin when making decisions about the management of hyperbilirubinaemia (see management thresholds and treatment in (threshold graphs)

7.1-Measuring and monitoring bilirubin thresholds during phototherapy

Starting phototherapy Use serum bilirubin measurement and the treatment thresholds in the threshold table and treatment threshold graphs

7.2-When considering the use of phototherapy.

- In babies with a gestational age of 38 weeks or more whose bilirubin is in the ‘consider phototherapy’ category in the threshold table repeat the bilirubin measurement in 6 hours regardless of whether or not phototherapy has subsequently been started.
- Do not use phototherapy in babies whose bilirubin does not exceed the phototherapy threshold levels in the threshold table and treatment threshold graphs

3-Type of phototherapy to use

Do not use sunlight as treatment for hyperbilirubinaemia.

7.3.1-Single phototherapy treatment for term babies

Use conventional ‘blue light’ phototherapy as treatment for significant hyperbilirubinaemia in babies with a gestational age of 38 weeks or more unless:

- the serum bilirubin level is rising rapidly (more than 8.5 micromol/litre per hour)
- the serum bilirubin is at a level that is within 50 micromol/litre below the threshold for which exchange transfusion is indicated after 72 hours (see the threshold table and treatment threshold graphs)

7.3.2-Single phototherapy treatment in preterm babies

Use conventional ‘blue light’ phototherapy as treatment for significant hyperbilirubinaemia in babies less than 38 weeks unless:

- the serum bilirubin level is rising rapidly (more than 8.5 micromol/litre per hour).
- the serum bilirubin is at a level that is within 50 micromol/litre below the threshold for which exchange transfusion is indicated after 72 hours (see threshold graphs)

7.3.3-Continuous multiple phototherapy treatment for term and pre term babies

Initiate continuous multiple phototherapy to treat all babies if any of the following apply:

- The serum bilirubin level is rising rapidly (more than 8.5 micromol/litre per hour).
- The serum bilirubin is at a level within 50 micromol/litre below the threshold for which exchange transfusion is indicated after 72 hours (see threshold table and treatment threshold graphs)
- The bilirubin level fails to respond to single phototherapy (that is, the level of serum bilirubin continues to rise, or does not fall, within 6 hours of starting single phototherapy).
- If the serum bilirubin level falls during continuous multiple phototherapy to a level 50 micromol/litre below the threshold for which exchange transfusion is indicated step down to single phototherapy.

7.3.4-Information for parents or carers on phototherapy

Offer parents or carers verbal and written information on phototherapy including all of the following:

- Reassurance that short breaks for feeding, nappy changing and cuddles will be encouraged.
- What might happen if phototherapy fails .
- rebound jaundice
- Potential impact on breastfeeding and how to minimize this.

7.3.5-General care of the baby during phototherapy

During phototherapy:

- Baby position according to nursing unit policy
- Ensure treatment is applied to the maximum area of skin.
- Monitor the baby's temperature and ensure the baby is kept in an environment that will minimize energy expenditure (thermo neutral environment).
- Monitor hydration by daily weighing of the baby and assessing wet nappies.
- Support parents and carers and encourage them to interact with the baby.
- Give the baby eye protection and routine eye care according to unite policy during phototherapy.

7.3.5-Monitoring the baby during phototherapy

7.3.5.1-During conventional 'blue light' phototherapy:

- Using clinical judgement, encourage short breaks (of up to 30 minutes) for breastfeeding, nappy changing and cuddles.
- Continue lactation/feeding support.
- Don't give additional fluids or feeds routinely.
- Maternal expressed milk is the additional feed of choice if available, and when additional feeds are indicated.

7.3.6-During phototherapy:

- Repeat serum bilirubin 4–6 hr. after starting treatment
- Repeat serum bilirubin 6–12 hr. when bilirubin stable or falling

7.3.7-Stopping phototherapy

- Stop phototherapy once serum bilirubin has fallen to a level at least 50 micromol/litre below the phototherapy threshold (see threshold table and treatment threshold graphs)
- Check for rebound of significant hyperbilirubinaemia with a repeat serum bilirubin measurement 12–18 hours after stopping phototherapy.
- Babies do not necessarily have to remain in hospital for this to be done.

8-DISCHARGE AND FOLLOW-UP

- In babies with more than weakly positive Coombs' test who require phototherapy:
- check haemoglobin at aged 2 and 4 weeks due to risk of continuing haemolysis
- give folic acid 1 mg daily

9-Factors that influence the risk of kernicterus I

- A serum bilirubin level greater than 340 micromol/litre in babies with a gestational age of 37 weeks or more.
- A rapidly rising bilirubin level of greater than 8.5 micromol/litre per hour.
- Clinical features of acute bilirubin encephalopathy.

10 -Intravenous immunoglobulin

10.1 Indication

- Rh and ABO incompatibility when the serum bilirubin continues to rise by more than 8.5 micromol/litre per hour.
- Not indicated unless Direct Antibody Test (DAT) is positive.
- Other isoimmune haemolytic disease (no systematic reviews available).
- Difficulties in obtaining appropriate blood for ET or parental refusal for ET.
- Do not give if exchange transfusion imminent

10.2-Dose and Administration

Privigen 10% (Immunoglobulin IgG) from pooled human plasma

Presentation: 2.5g in 50mL vial. 0.5 to 1g/kg as an Intravenous infusion over 4 hours

10.3-Side Effects

Possible side effects are similar to blood transfusions.

- Fever, allergic reactions, haemolysis, fluid overload, anaphylaxis (reported in IgA deficiency).
- Possible disease transmission. Systematic reviews did not reveal any adverse reactions in neonates receiving IVIg.

10.4-Offer parents or carers information on IVIG including:

- Why IVIG is being considered.
- the possible adverse effects of IVIG
- When it will be possible for parents or carers to see and hold the baby.

11-Exchange transfusion

11.1-Offer parents or carers information on exchange transfusion including:

- The fact that exchange transfusion requires that the baby be admitted to an intensive care bed.
- Why an exchange transfusion is being considered.
- The possible adverse effects of exchange transfusions.
- When it will be possible for parents or carers to see and hold the baby after the exchange transfusion.

11.2- Use a double-volume exchange transfusion to treat babies:

- Or with clinical features and signs of acute bilirubin encephalopathy.

11.3-During exchange transfusion

- Don't stop continuous multiple phototherapy.
- Don't perform a single-volume exchange.
- Don't use albumin priming.
- Don't routinely administer intravenous calcium.

11.4-Following exchange transfusion:

- Maintain continuous multiple phototherapy.

measure serum bilirubin level within 2 hours and manage according to (threshold graphs).

12-References

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